ADEMP protocol: 08-test-procedure-design-sensitivity

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This document is the working protocol for the exploratory simulation study that underlies `analysis/report/08-test-procedure-design-sensitivity/`. The study is exploratory, and its findings should be confirmed in an independent study. ADEMP structure follows Morris, White and Crowther (2019).

Common infrastructure

Trial designs. Hendrickson hybrid OL+BDC as the reference, plus three trial-design variants on the cycle-by-period grid (see Study 2 below).

Sample sizes. $N = 35$ per design at the baseline, varied across $\{35, 70, 100\}$ in the N-axis sweep.

Drug context. Prazosin for PTSD nightmares; baseline parameters at the Hendrickson 2020 reference values.

Carryover. Exponential decay with $t_{1/2} = 3$ days at the baseline cell.

Replicates. $n_{\text{reps}} = 1000$ per cell for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.

Random-number control. Per-replicate seed via cell-descriptor hash; base seed in `01-base-seed.txt`.

Reporting standard. Morris-White-Crowther ADEMP with MCSE columns for every reported performance measure.

Simulation substrate. `implementations/simple/simulation.R` extended with the cycle-by-period axes (rather than the full pmsimstats package stack), per the substrate-choice argument in §1 of the manuscript.

Study 1. Test-procedure comparison at fixed design

A. Quantify the relative power, type I error, and bias of three test procedures (strict classical RM-ANOVA, linear mixed- effects with corCAR1 residual correlation, GEE with bias- corrected sandwich variance) for the biomarker-treatment interaction, holding the trial-design parameters at the Hendrickson hybrid reference.

D. Three-component additive DGP at the prazosin baseline, biomarker effect $c_{bm} \in \{0, 0.3, 0.6\}$, with the design fixed at the Hendrickson hybrid (8-week OL + 4-week BDC + 4-week + 4-week crossover blocks).

E. Primary estimand: $\hat{\beta}_{bm:D}$ from the analysis- model fixed-effects table for the linear-mixed and GEE analyses; the corresponding F-statistic from the strict RM-ANOVA. The p-value associated with each is the inferential output that drives power and type I error.

M. Three analysis variants compared:

1. *Strict classical RM-ANOVA* per Greenhouse-Geisser-corrected F-statistic on the biomarker-stratified treatment-by-time interaction in a balanced split-plot design.
2. *Linear mixed-effects with corCAR1*: `Sx ~ bm + t + Dbc + bm:Dbc + (1 | ptID)` with `corCAR1(form = ~t|ptID)` (the pmsimstats default analysis).
3. *GEE with bias-corrected sandwich variance* per @wang2016geesmv: `geeglm(Sx ~ bm + t + Dbc + bm:Dbc, id = ptID, corstr = 'ar1')` with the Mancl-DeRouen correction.

P. Power, type I error, bias, empirical SE, model SE, nominal-95% CI coverage, convergence rate. MCSE attached to every estimate. Coverage is computed for the linear-mixed and GEE analyses (where intervals are produced); for RM-ANOVA only the rejection-rate quantities are reported.

Working prediction. Linear-mixed dominates GEE on power at trial-relevant N due to the small-sample bias of the sandwich estimator, even with correction; both dominate strict RM-ANOVA when within-participant residuals depart from compound symmetry, which is the empirically common case.

Study 2. Cycle-by-period design grid at fixed test procedure

A. Quantify the dependence of the biomarker-treatment interaction test on the cycle-by-period design grid, holding the test procedure at the linear-mixed-with-corCAR1 baseline.

D. Three-component DGP at the prazosin baseline, with the design parameters varied across:

- Number of treatment cycles $k \in \{1, 2, 3, 4, 6\}$.
- On-drug period length $L_{\text{on}} \in \{1, 2, 3, 4\}$ weeks.
- Off-drug period length $L_{\text{off}} \in \{1, 2, 3, 4\}$ weeks.
- On/off symmetry indicator (factorialised as $L_{\text{on}} = L_{\text{off}}$ vs $\neq$).

E. $\hat{\beta}_{bm:D}$ from the linear-mixed analysis; power and bias as a function of the design grid.

M. Linear-mixed-with-corCAR1 analysis only.

P. Power, type I error, bias, MCSE, convergence rate, mean fit time. The fit-time measure characterises the computational cost of long-cycle designs.

Working prediction. Per-participant power is monotone in cycle count k with diminishing returns above $k = 4$. On-drug period length matters more for power than off-drug period length when carryover is non-trivial. Asymmetric $L_{\text{on}} > L_{\text{off}}$ splits are preferable when carryover is non-trivial.

Study 3. Joint design-by-test heatmap

A. Synthesise Studies 1 and 2 into a joint design-by-test recommendation table.

D. Same DGP grid as Study 2, with the three test procedures from Study 1 each fitted to every cell.

E. Power and type I error for each (design, test) combination.

M. As Studies 1 and 2.

P. A heatmap of power as a function of design parameters crossed with test procedure, with MCSE annotation on every cell.

Total cell counts

- Study 1: 3 tests $\times$ $3N \times 3c_{bm} = 27$ cells.
- Study 2: $5k \times 4L_{\text{on}} \times 4L_{\text{off}} \times 3c_{bm} = 240$ cells.
- Study 3: subset of Studies 1-2 with all three tests applied: 3 tests $\times$ $5k \times 4L_{\text{on}} \times 4L_{\text{off}} = 240$ cells.

Deliverables

- Driver scripts at `analysis/scripts/test-procedure-design-sensitivity/`.
- ADEMP results tables per study with **MCSE columns alongside every estimate**, per Morris recommendation.
- Figures: test-procedure comparison plots faceted by N and c_{bm}; design-grid heatmaps; joint design-by-test heatmap.

Protocol change log

`02-deviations-log.md` records changes to the cells, estimands, methods, or performance measures specified in this working protocol.

Source: `~/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/scripts/test-procedure-design-sensi`